

EXPERIMENT - 4

Date: 20.8.2016

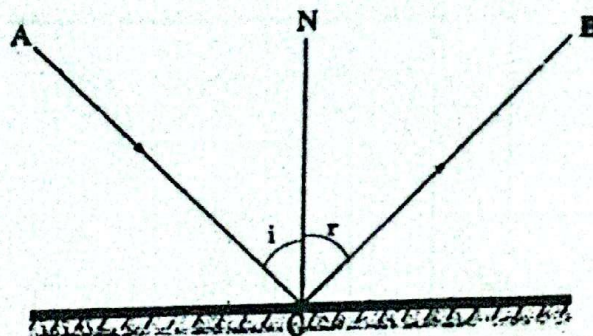
TO VERIFY THE LAWS OF REFLECTION OF LIGHT**APPARATUS REQUIRED:**

- Plane mirror mounted on wooden block
- A sheet of paper
- Drawing board
- Hair pins & fixing pins
- Scale
- Protractor

THEORY:

The laws of reflection of light states that

- The angle of incidence is equal to the angle of reflection.
- The incident ray, normal at the point of incidence and the reflected ray all lie in the same plane.

**OBSERVATIONS:**

L.C. of protector = 1°

L.C. of scale = 0.1 cm

Observation table I: For the verification of the first law of reflection (Using Protractor)

No. of Obs.	Angle of incidence I	Angle of reflection R	Remarks
1	20°	20.1°	$\angle i \approx \angle R$
2	30°	30.1°	$\angle i \approx \angle R$
3	40°	40.1°	$\angle i \approx \angle R$
4	50°	50.1°	$\angle i \approx \angle R$

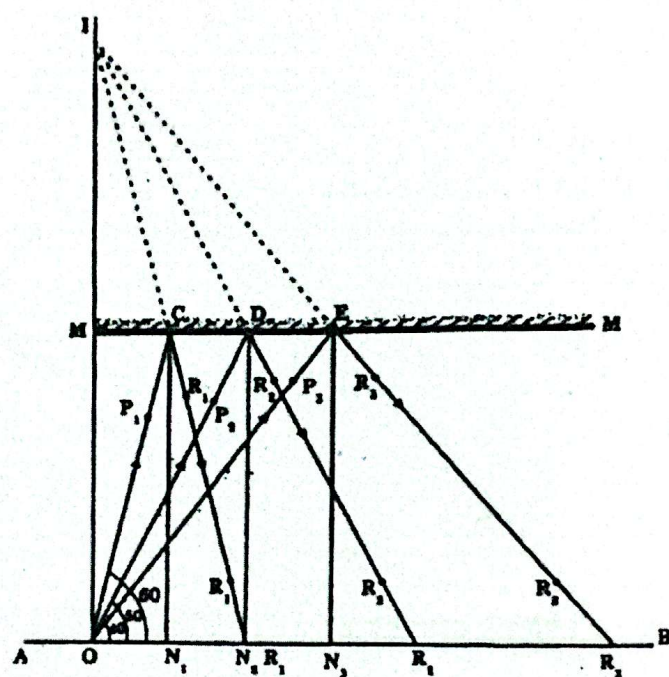
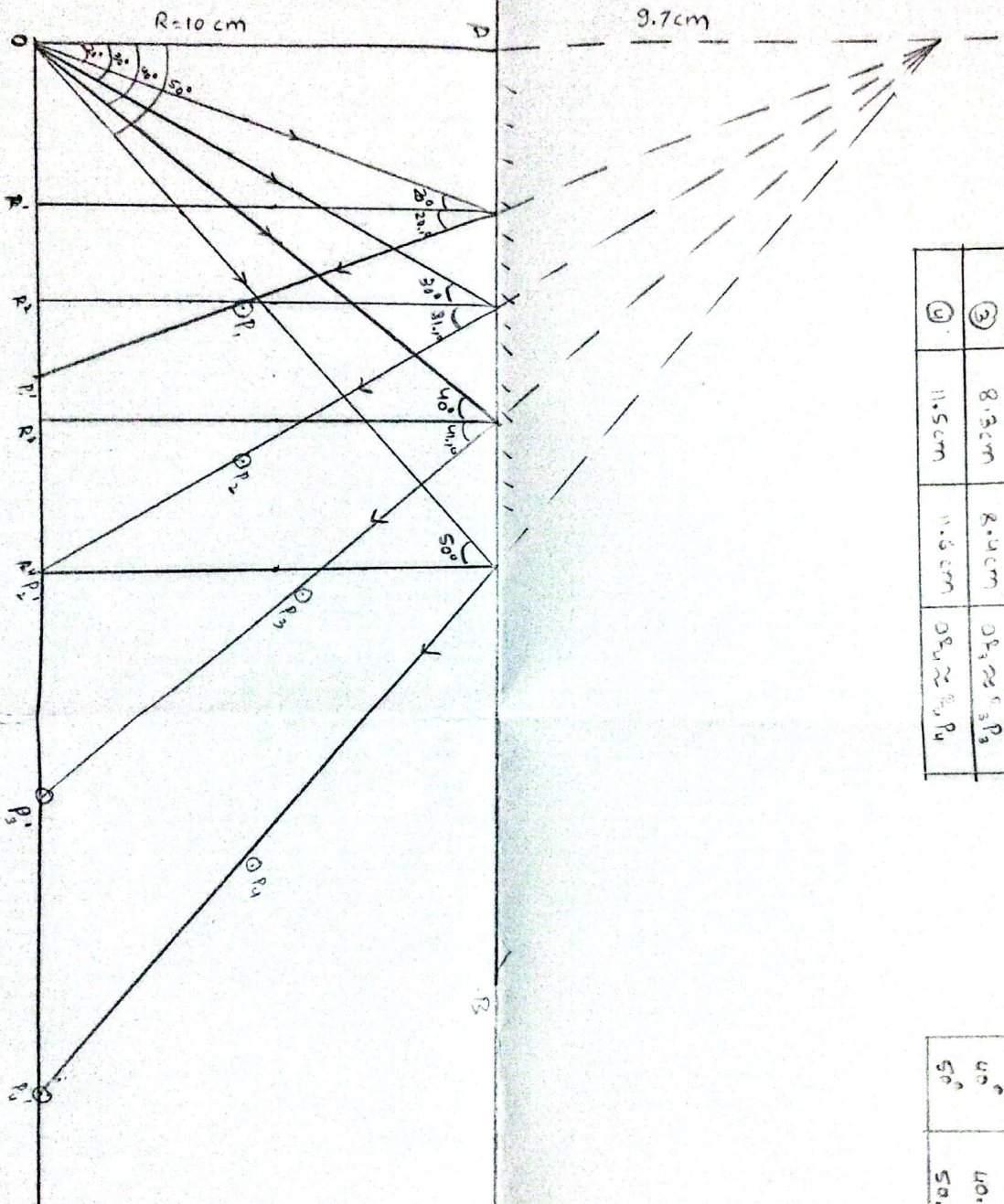


Fig. Reflection of Light

Without protactor

	θ_R	θ_P	Remarks
①	3.5 cm	3.8 cm	$\theta_R \approx \theta_P, P_1$
②	5.7 cm	6 cm	$\theta_R \approx \theta_P, P_2$
③	8.3 cm	8.4 cm	$\theta_R \approx \theta_P, P_3$
④	11.5 cm	11.6 cm	$\theta_R \approx \theta_P, P_4$

Δ_i	Δ_R	Remarks
20°	20.1°	$\Delta_i \approx \Delta_R$
30°	20.1°	$\Delta_i \approx \Delta_R$
40°	40.1°	$\Delta_i \approx \Delta_R$
50°	50.1°	$\Delta_i \approx \Delta_R$



Observation table II: For the verification of the first law of reflection (without using protractor)

No. of Obs.	Distance (ON) cm	Distance (NR) cm	Remarks
1	3.5 cm	3.8 cm	$ON_1 \approx N_1R_1$
2	5.7 cm	6 cm	$ON_2 \approx N_2R_2$
3	8.3 cm	8.4 cm	$ON_3 \approx N_3R_3$
4	11.5 cm	11.6 cm	$ON_4 \approx N_4R_4$

Object Distance (u) = 10 cm

Image Distance (v) = 9.7 cm

RESULT:

Hence, from this experiment, we verified the law of reflection (first and second law)

CONCLUSION:

From this experiment, we verified the law of reflection using - plane mirror.

SOURCES OF ERROR:

The mirror may not be exactly perpendicular to the plane. the object pin and image may not lie in the same line while viewing the mirror. The two image pin may-be too near to each other.

PRECAUTIONS:

- 1) The mirror should be exactly perpendicular to the plane of mirror.
- 2) The polished surface of the mirror should coincide with line XY.
- 3) The separation between the pins should be about 10 cm.

VIVA QUESTIONS:

1. What do you meant by reflection of light? State the laws.
2. What is glancing angle?
3. Is it possible to obtain a real image by a plane mirror?
4. What do you mean by regular and diffuse reflection?
5. Does the law of reflection of light hold good in diffused reflection?

6. What is the focal length of plane mirror?
7. Distinguish between real and virtual image.
8. Why a plane mirror can show your image but a paper cannot?

① The phenomenon of returning back of light in same medium when light incident on a surface of a medium is called reflection of light.

The laws are:

- a) The angle of incidence is equal to the angle of reflection. i.e. $\angle i = \angle r$
- b) The incident rays, the reflected ray and the normal all lies in a same point of same plane.

② The angle made by incident or regular reflected ray with the polished surface is called glancing angle.

③ Yes, a plane mirror can form a real image. This happens when the object is virtual and image is real.

④ The reflection from a smooth plane surface is regular reflection. Where as, the reflection from a rough surface is irregular reflection.

⑤ Yes, the law of reflection of light holds good in diffused reflection.

⑥ For the plane mirror, the radius of curvature is infinite and from the relation,

$$F = \frac{R}{2} \text{ so,}$$

the focal length of plane mirror is infinite i.e. $f = \infty$.

⑦ The differences of real and virtual image are:-

Real Image	Virtual Image
- It can be obtained on a screen.	It cannot be obtained on a screen.
- Real image is inverted.	Virtual image is erected.
- When two reflected or refracted rays are directly intersected then, a real image is formed.	When two reflected or refracted rays are intersected by producing backward, is as then virtual image is formed.
- E.g - Image formed by camera.	e.g - Image formed by plane mirror.

⑧ A plane mirror can show our image but a paper cannot because a plane mirror has a reflecting surface whereas a paper does not.